WORKED SOLUTION

STEP I 1988, Question 4

Pure • Elementary geometry

> **STEP I 1988, Question 4**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/88-s1-q4

The key idea

Don't try to picture all the segments at once. Ask instead: when do *two* segments cross? The answer depends only on the order of their endpoints along the axes, and it turns every crossing into a choice of two points on each axis. Once we can count the crossings on a single segment, the extreme segments and the final average follow by careful bookkeeping.

Solution

Label the points on the y-axis $B_1, \ldots, B_m$ in order of increasing distance from the origin O, and the points on the x-axis $A_1, \ldots, A_n$ in the same way. Write s_{ij} for the segment B_iA_j.

Part (i)

Each of the m points B_i is joined to each of the n points A_j, so there are

$$mn$$

segments.

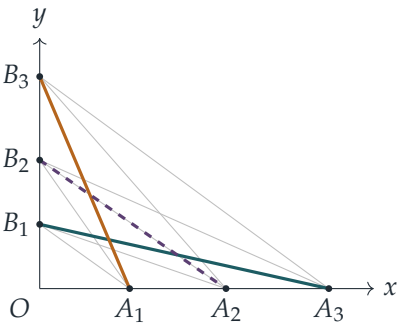
Part (ii)

Take two different segments s_{ij} and s_{kl}.

- If $i = k$ or $j = l$, they share an endpoint. Two distinct segments from a common point meet only there, and those meetings are ignored.
- If $i < k$ and $j < l$, both B_i and A_j lie on the same side of the line B_kA_l as O (they are on the legs OB_k and OA_l of the triangle OB_kA_l, not at B_k or A_l). A segment whose endpoints lie strictly on one side of a line stays on that side, so s_{ij} never meets s_{kl}: it is "nested" underneath.

- If $i < k$ and $j > l$, then B_i is on the same side of the line B_kA_l as O, while A_j is on the far side. So s_{ij} crosses that line. In the same way, A_l is on the O side of the line B_iA_j and B_k is on the far side. Each segment separates the endpoints of the other, so the two segments cross at exactly one point, which is not an endpoint.

The case $i > k$ is the same with the roles swapped.



In the figure ($m = n = 3$), the teal s_{13} and amber s_{31} have their endpoints in opposite orders, so they cross. The dashed s_{22} crosses each of them too.

So each pair of segments crossing at an interior point comes from choosing two points B_i, B_k ($i < k$) and two points A_j, A_l ($j < l$), and then taking s_{il} and s_{kj}. Conversely, each such choice gives exactly one crossing pair. Since no three segments pass through a common interior point, different pairs give different points. Hence the number of points of intersection is

$$\binom{m}{2}\binom{n}{2} = \frac{m(m-1)n(n-1)}{4}.$$

The average

Crossings on one segment. By the rule above, s_{ij} crosses s_{kl} exactly when $k < i,\ l > j$ or $k > i,\ l < j$. So the number of points of intersection on s_{ij} is

$$c_{ij} = (i-1)(n-j) + (m-i)(j-1).$$

The extreme segments when $m = n$. Put $N = n-1$, $a = i-1$, $b = j-1$, so $0 \le a, b \le N$ and

$$c_{ij} = a(N-b) + (N-a)b = N(a+b) - 2ab = \tfrac{1}{2}\big(N^2 - (N-2a)(N-2b)\big),$$

as you can check by expanding. Since $|N-2a| \le N$ and $|N-2b| \le N$, the product $(N-2a)(N-2b)$ lies between $-N^2$ and N^2. Therefore $0 \le c_{ij} \le N^2$, and:

- $c_{ij} = N^2$ only when $(N-2a)(N-2b) = -N^2$, that is, $\{a, b\} = \{0, N\}$. These are the two segments s_{1n} and s_{n1}.
- $c_{ij} = 0$ only when $(N-2a)(N-2b) = N^2$, that is, $a = b = 0$ or $a = b = N$. These are the two segments s_{11} and s_{nn}.

(Here $N \geq 2$, so the four segments are distinct.) So exactly two segments carry the greatest number, $(n-1)^2$, of points, and exactly two carry the least, 0. Removing them leaves $n^2 - 4$ segments.

Ask yourself What exactly is being averaged?

We count the points of intersection that remain once the four segments are gone, i.e. crossings between two remaining segments, each point counted once. Then we divide by the number of remaining segments. (If you instead add up the crossings on each remaining segment, every point is counted twice and some lost points are still counted. That gives $\frac{1}{2}(n-1)^2$, which is not the stated answer. See Common slips.)

Points lost. The segments s_{11} and s_{nn} carry no points of intersection, so removing them loses nothing. The segments s_{1n} and s_{n1} carry $(n-1)^2$ points each. They cross each other ($1 < n$ and $n > 1$), so exactly one point lies on both. By inclusion–exclusion, the number of points lost is $2(n-1)^2 - 1$.

Points remaining. From part (ii) with $m = n$, the number of points remaining is

$$\begin{aligned}\frac{n^2(n-1)^2}{4} - 2(n-1)^2 + 1 &= \frac{(n-1)^2(n^2-8)+4}{4} \\ &= \frac{n^4 - 2n^3 - 7n^2 + 16n - 4}{4} \\ &= \frac{(n-2)(n^3 - 7n + 2)}{4}.\end{aligned}$$

For the factorisation, note that $n = 2$ makes the quartic $16 - 16 - 28 + 32 - 4 = 0$, so $n - 2$ is a factor. Multiplying out $(n-2)(n^3 - 7n + 2) = n^4 - 2n^3 - 7n^2 + 16n - 4$ confirms it.

The average. Dividing by the $n^2 - 4 = (n-2)(n+2)$ remaining segments, and cancelling $n - 2 \neq 0$ (since $n \geq 3$), gives an average of

$$\frac{(n-2)(n^3 - 7n + 2)}{4(n-2)(n+2)} = \frac{n^3 - 7n + 2}{4(n+2)},$$

as required.

Checks

For $n = 3$ the formula gives $\frac{8}{20} = \frac{2}{5}$. Directly: removing $s_{11}, s_{33}, s_{13}, s_{31}$ leaves $s_{12}, s_{21}, s_{22}, s_{23}, s_{32}$. Among these, only s_{12} with s_{21}, and s_{23} with s_{32}, have endpoints in opposite orders, so there are 2 points on 5 segments, which gives $\frac{2}{5}$. A computer check placed random points on the axes, found every crossing with exact arithmetic, and confirmed mn, $\binom{m}{2}\binom{n}{2}$ and c_{ij}. It also confirmed the average formula for every n from 3 to 29.

Common slips

- Adding the crossings on each remaining segment. Each point lies on two segments, so this double counts, and it still includes points shared with the removed segments. The result is $\frac{1}{2}(n-1)^2$, not the given formula.
- Forgetting that s_{1n} and s_{n1} cross each other, and so subtracting $2(n-1)^2$ instead of $2(n-1)^2-1$.
- Counting crossings as $\binom{mn}{2}$ pairs of segments: most pairs share an endpoint or are nested and never cross.

Beyond STEP

The heart of this question is a bijection: crossings correspond exactly to choices of two points on each axis. This is the basic tool of combinatorics. The same picture appears in the "crossing number" of the complete bipartite graph $K_{m,n}$. Zarankiewicz's drawing (exactly this one, with points on both sides of each axis) uses $\lfloor \frac{m}{2} \rfloor \lfloor \frac{m-1}{2} \rfloor \lfloor \frac{n}{2} \rfloor \lfloor \frac{n-1}{2} \rfloor$ crossings, and whether this is always the least possible is still an open problem. For more counting by bijection, try Siklos, *Advanced Problems in Mathematics*, and for graph theory, Wilson, *Introduction to Graph Theory*.

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